1.02 Surds

Worked Solutions

1 a)

$\sqrt{32} = \sqrt{16 \times 2} = 4\sqrt{2}$

1 b)

$2\sqrt{12} = 2\sqrt{4 \times 3} = 4\sqrt{3}$

1 c)

$\sqrt{45} = \sqrt{9 \times 5} = 3\sqrt{5}$

1 d)

$\sqrt{200} = \sqrt{100 \times 2} = 10\sqrt{2}$

1 e)

$-4\sqrt{50} = -4\sqrt{25 \times 2} = -20\sqrt{2}$

2 a)

$\sqrt{5}\sqrt{2} = \sqrt{10}$

2 b)

$\sqrt{5}\sqrt{15} = \sqrt{75} = \sqrt{25 \times 3} = 5\sqrt{3}$

2 c)

$3\sqrt{2}\sqrt{8} = 3\sqrt{16} = 12$

2 d)

$\sqrt{\frac{1}{9}} = \frac{\sqrt{1}}{\sqrt{9}} = \frac{1}{3}$

2 e)

$\frac{\sqrt{28}}{\sqrt{14}} = \sqrt{\frac{28}{14}} = \sqrt{2}$

2 f)

$\sqrt{25x^2} = \sqrt{25}\sqrt{x^2} = 5x$

3 a)

$2\sqrt{5} - \sqrt{5} = \sqrt{5}$

3 b)

$\sqrt{8} + 2\sqrt{2} = \sqrt{4 \times 2} + 2\sqrt{2} = 2\sqrt{2} + 2\sqrt{2} = 4\sqrt{2}$

3 c)

$\sqrt{12} - \sqrt{3} = \sqrt{4 \times 3} - \sqrt{3} = 2\sqrt{3} - \sqrt{3} = \sqrt{3}$

3 d)

$\sqrt{128} - \sqrt{72} = \sqrt{64 \times 2} - \sqrt{36 \times 2} = 8\sqrt{2} - 6\sqrt{2} = 2\sqrt{2}$

3 e)

$3\sqrt{12} + \sqrt{27} = 3\sqrt{4 \times 3} + \sqrt{9 \times 3} = 6\sqrt{3} + 3\sqrt{3} = 9\sqrt{3}$

3 f)

$\sqrt{150} - \sqrt{96} = \sqrt{25 \times 6} - \sqrt{16 \times 6} = 5\sqrt{6} - 4\sqrt{6} = \sqrt{6}$

3 g)

$\sqrt{3} + \sqrt{27} + \sqrt{12} = \sqrt{3} + \sqrt{9 \times 3} + \sqrt{4 \times 3} = \sqrt{3} + 3\sqrt{3} + 2\sqrt{3} = 6\sqrt{3}$

3 h)

$\sqrt{350} + \sqrt{24} = \sqrt{25 \times 14} + \sqrt{4 \times 6} = 5\sqrt{14} + 2\sqrt{6}$

3 i)

$\sqrt{32a} + 2\sqrt{a} - \sqrt{28a} = \sqrt{16 \times 2a} + 2\sqrt{a} - \sqrt{4 \times 7a} = 4\sqrt{2a} + 2\sqrt{a} - 2\sqrt{7a}$

4 a)

$\sqrt{2} \times \sqrt{7} \times \sqrt{3} = \sqrt{42}$

4 b)

$\sqrt{18} \times \sqrt{2} = \sqrt{36} = 6$

4 c)

$2\sqrt{3} \times 2\sqrt{3} = 4 \times 3 = 12$

4 d)

$5\sqrt{3} \times 2\sqrt{5} = 10\sqrt{15}$

4 e)

$-3\sqrt{3} \times 2\sqrt{3} = -6 \times 3 = -18$

4 f)

$(2\sqrt{6})^2 = 4 \times 6 = 24$

4 g)

$(3\sqrt{8})^2 = 9 \times 8 = 72$

4 h)

$(2\sqrt{5})^3 = 8 \times 5\sqrt{5} = 40\sqrt{5}$

5 a)

$2\sqrt{5} = \sqrt{2^2 \times 5} = \sqrt{20}$

5 b)

$4\sqrt{3} = \sqrt{4^2 \times 3} = \sqrt{48}$

5 c)

$3\sqrt{2} = \sqrt{3^2 \times 2} = \sqrt{18}$

5 d)

$2\sqrt{7} = \sqrt{2^2 \times 7} = \sqrt{28}$

5 e)

$2\sqrt{43} = \sqrt{2^2 \times 43} = \sqrt{172}$

6 a)

$a\sqrt{b} = \sqrt{a^2 b}$

6 b)

$2\sqrt{a} = \sqrt{2^2 a} = \sqrt{4a}$

6 c)

$(\sqrt{a})^5 = \sqrt{a} \times \sqrt{a} \times \sqrt{a} \times \sqrt{a} \times \sqrt{a} = a \times a \times \sqrt{a} = a^2\sqrt{a} = \sqrt{a^4 \times a} = \sqrt{a^5}$

6 d)

$(a\sqrt{b})^3 = a\sqrt{b} \times a\sqrt{b} \times a\sqrt{b} = a^3 b\sqrt{b} = \sqrt{a^6 b^2 b} = \sqrt{a^6 b^3}$

7)

$\sqrt{525} = \sqrt{25 \times 3 \times 7} = 5\sqrt{3}\sqrt{7}$

8)

$\sqrt{100 \times 3} : \sqrt{16 \times 3} : \sqrt{9 \times 3}$

$10\sqrt{3} : 4\sqrt{3} : 3\sqrt{3}$

$10 : 4 : 3$

9 a)

$\frac{3}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$

9 b)

$\frac{6}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{6\sqrt{3}}{3} = 2\sqrt{3}$

9 c)

$\frac{6}{\sqrt{10}} \times \frac{\sqrt{10}}{\sqrt{10}} = \frac{6\sqrt{10}}{10} = \frac{3\sqrt{10}}{5}$

9 d)

$\frac{2}{\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}} = \frac{2\sqrt{6}}{6} = \frac{\sqrt{6}}{3}$

9 e)

$\frac{10}{\sqrt{10}} \times \frac{\sqrt{10}}{\sqrt{10}} = \frac{10\sqrt{10}}{10} = \sqrt{10}$

9 f)

$\frac{4\sqrt{2}}{\sqrt{8}} \times \frac{\sqrt{8}}{\sqrt{8}} = \frac{4\sqrt{16}}{8} = \frac{16}{8} = 2$

10 a)

$\sqrt{5}(1+\sqrt{5}) = \sqrt{5} + \sqrt{5}\sqrt{5} = \sqrt{5} + 5$

10 b)

$\sqrt{11}(2+\sqrt{11}) = 2\sqrt{11} + \sqrt{11}\sqrt{11} = 2\sqrt{11} + 11$

10 c)

$4(\sqrt{3}+2\sqrt{5}) = 4\sqrt{3} + 8\sqrt{5}$

10 d)

$(\sqrt{3}+1)(\sqrt{3}-1) = \sqrt{3}\sqrt{3} - \sqrt{3} + \sqrt{3} - 1 = 3 - 1 = 2$

10 e)

$(\sqrt{6}+1)(2+\sqrt{6}) = 2\sqrt{6} + \sqrt{6}\sqrt{6} + 2 + \sqrt{6} = 3\sqrt{6} + 6 + 2 = 3\sqrt{6} + 8$

10 f)

$(\sqrt{8}-1)(2+\sqrt{8}) = 2\sqrt{8} + \sqrt{8}\sqrt{8} - 2 - \sqrt{8} = \sqrt{8} + 8 - 2 = \sqrt{8} + 6 = 2\sqrt{2} + 6$

10 g)

$$\begin{aligned}(\sqrt{24}+1)(\sqrt{2}-3) &= \sqrt{24}\sqrt{2} - 3\sqrt{24} + \sqrt{2} - 3 \\ &= \sqrt{16\times 3} - 3\sqrt{4\times 6} + \sqrt{2} - 3 \\ &= 4\sqrt{3} - 6\sqrt{6} + \sqrt{2} - 3\end{aligned}$$

11 a)

$\frac{1}{\sqrt{2}+1} \times \frac{\sqrt{2}-1}{\sqrt{2}-1} = \frac{\sqrt{2}-1}{2-1} = \sqrt{2} - 1$

11 b)

$\frac{3}{\sqrt{5}-2} \times \frac{\sqrt{5}+2}{\sqrt{5}+2} = \frac{3(\sqrt{5}+2)}{5-4} = 3\sqrt{5} + 6$

11 c)

$\frac{2}{1-\sqrt{6}} \times \frac{1+\sqrt{6}}{1+\sqrt{6}} = \frac{2+2\sqrt{6}}{1-6} = -\frac{2}{5} - \frac{2}{5}\sqrt{6}$

11 d)

$\frac{1}{a+\sqrt{b}} \times \frac{a-\sqrt{b}}{a-\sqrt{b}} = \frac{a-\sqrt{b}}{a^2-b}$

11 e)

$\frac{1}{a-\sqrt{2b}} \times \frac{a+\sqrt{2b}}{a+\sqrt{2b}} = \frac{a+\sqrt{2b}}{a^2-2b}$

12 a)

$\sqrt{27} + \frac{15}{\sqrt{3}} = \sqrt{9 \times 3} + \frac{15}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = 3\sqrt{3} + 5\sqrt{3} = 8\sqrt{3}$

12 b)

$\frac{\sqrt{54}}{2} - \frac{10}{\sqrt{6}} = \frac{\sqrt{9\times6}}{2} - \frac{10}{\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}} = \frac{3}{2}\sqrt{6} - \frac{10}{6}\sqrt{6} = -\frac{1}{6}\sqrt{6}$

12 c)

$$\begin{aligned}\frac{\sqrt{150}}{5} + \frac{30}{\sqrt{24}} &= \frac{\sqrt{25 \times 6}}{5} + \frac{30}{\sqrt{24}} \times \frac{\sqrt{24}}{\sqrt{24}} \\ &= \sqrt{6} + \frac{30}{24}\sqrt{24} = \sqrt{6} + \frac{5}{4}\sqrt{4 \times 6} \\ &= \sqrt{6} + \frac{5}{2}\sqrt{6} = \frac{7}{2}\sqrt{6}\end{aligned}$$

13)

$x = \frac{10}{4-\sqrt{6}} \times \frac{4+\sqrt{6}}{4+\sqrt{6}} = \frac{40+10\sqrt{6}}{16-6} = 4 + \sqrt{6}$

14 a)

$\frac{\sqrt{3}}{\sqrt{2}-2} \times \frac{\sqrt{2}+2}{\sqrt{2}+2} = \frac{\sqrt{3}\sqrt{2}+2\sqrt{3}}{2-4} = \frac{\sqrt{6}+2\sqrt{3}}{-2} = -\frac{1}{2}\sqrt{6} - \sqrt{3}$

14 b)

$\frac{2\sqrt{5}-1}{\sqrt{5}+4} \times \frac{\sqrt{5}-4}{\sqrt{5}-4} = \frac{10-8\sqrt{5}-\sqrt{5}+4}{5-16} = \frac{14-9\sqrt{5}}{-11} = -\frac{14}{11} + \frac{9}{11}\sqrt{5}$

14 c)

$$\begin{aligned}\frac{5\sqrt{3}-2}{2-\sqrt{7}} \times \frac{2+\sqrt{7}}{2+\sqrt{7}} &= \frac{10\sqrt{3}+5\sqrt{21}-4-2\sqrt{7}}{4-7} \\ &= \frac{10\sqrt{3}+5\sqrt{21}-4-2\sqrt{7}}{-3} \\ &= -\frac{10\sqrt{3}}{3} - \frac{5\sqrt{21}}{3} + \frac{4}{3} + \frac{2\sqrt{7}}{3}\end{aligned}$$

14 d)

$\frac{2\sqrt{3}-7}{2\sqrt{3}-3} \times \frac{2\sqrt{3}+3}{2\sqrt{3}+3} = \frac{12+6\sqrt{3}-14\sqrt{3}-21}{12-9} = \frac{-9-8\sqrt{3}}{3} = -3 - \frac{8}{3}\sqrt{3}$

14 e)

$$\begin{aligned}\frac{1-\sqrt{k}}{2+\sqrt{2k}} \times \frac{2-\sqrt{2k}}{2-\sqrt{2k}} &= \frac{2-\sqrt{2k}-2\sqrt{k}+\sqrt{k}\sqrt{2k}}{4-2k} \\ &= \frac{2-(\sqrt{2}+2)\sqrt{k}+\sqrt{2}k}{4-2k}\end{aligned}$$

15)

$x = \frac{5-\sqrt{2}}{1-\sqrt{2}} \times \frac{1+\sqrt{2}}{1+\sqrt{2}} = \frac{5+5\sqrt{2}-\sqrt{2}-2}{1-2} = \frac{3+4\sqrt{2}}{-1} = -3 - 4\sqrt{2}$

16)

$\frac{3\sqrt{10}}{\sqrt{2}+\sqrt{5}} \times \frac{\sqrt{2}-\sqrt{5}}{\sqrt{2}-\sqrt{5}} = \frac{3\sqrt{20}-3\sqrt{50}}{2-5} = \frac{6\sqrt{5}-15\sqrt{2}}{-3} = 5\sqrt{2} - 2\sqrt{5}$

17)

$\frac{\sqrt{3}+\sqrt{7}}{\sqrt{3}-\sqrt{7}} \times \frac{\sqrt{3}+\sqrt{7}}{\sqrt{3}+\sqrt{7}} = \frac{3+2\sqrt{21}+7}{3-7} = \frac{10+2\sqrt{21}}{-4} = -\frac{5}{2} - \frac{1}{2}\sqrt{21}$

18)

$$\begin{aligned}\frac{a+\sqrt{6}}{1-\sqrt{6}} \times \frac{1+\sqrt{6}}{1+\sqrt{6}} &= \frac{a+a\sqrt{6}+\sqrt{6}+6}{1-6} \\ &= \frac{a+6+(a+1)\sqrt{6}}{-5} \\ &= -\frac{a+6}{5} - \frac{a+1}{5}\sqrt{6}\end{aligned}$$

so $p = -\frac{a+6}{5}$ and $q = -\frac{a+1}{5}$.

19)

$\frac{12}{\frac{1}{\sqrt{3}}+\sqrt{3}} = \frac{12\sqrt{3}}{1+3} = 3\sqrt{3}$

20)

$$\left(\frac{1}{\sqrt{3}-2}\right)^2 = \frac{1}{\sqrt{3}-2} \times \frac{1}{\sqrt{3}-2} = \frac{1}{3-4\sqrt{3}+4} = \frac{1}{7-4\sqrt{3}}$$

$$\begin{aligned}\pi\left(\frac{1}{\sqrt{3}-2}\right)^2(\sqrt{3}+1) &= \pi\frac{\sqrt{3}+1}{7-4\sqrt{3}} \times \frac{7+4\sqrt{3}}{7+4\sqrt{3}} \\ &= \pi\frac{7\sqrt{3}+12+7+4\sqrt{3}}{49-48} \\ &= \pi(19+11\sqrt{3}) \text{ as req.}\end{aligned}$$

21)

$$\begin{aligned}\frac{-k+2\sqrt{3}}{\sqrt{27}+1}\times\frac{\sqrt{27}-1}{\sqrt{27}-1} &= \frac{-\sqrt{27}k+k+18-2\sqrt{3}}{27-1}\\ &= \frac{-3\sqrt{3}k-2\sqrt{3}+k+18}{26}\\ &= \left(\frac{-3k-2}{26}\right)\sqrt{3}+\frac{k+18}{26}\end{aligned}$$

So $a=\frac{k+18}{26}$ and $b=\frac{-3k-2}{26}$.

22)

$\frac{2-\sqrt{k}}{1+\sqrt{k}}\times\frac{1-\sqrt{k}}{1-\sqrt{k}}=\frac{2-2\sqrt{k}-\sqrt{k}+k}{1-k}=\frac{k+2}{1-k}-\frac{3}{1-k}\sqrt{k}$
So $a=\frac{k+2}{1-k}$ and $b=-\frac{3}{1-k}$.

23 a)

$$\begin{aligned}\frac{\sqrt{a+1}-\sqrt{a}}{\sqrt{a+1}+\sqrt{a}}\times\frac{\sqrt{a+1}-\sqrt{a}}{\sqrt{a+1}-\sqrt{a}} &= \frac{(a+1)-\sqrt{a}\sqrt{a+1}-\sqrt{a}\sqrt{a+1}+a}{(a+1)-a}\\ &= \frac{2a+1-2\sqrt{a}\sqrt{a+1}}{1}\\ &= 2a+1-2\sqrt{a}\sqrt{a+1}\end{aligned}$$

23 b)

So $\frac{\sqrt{6+1}-\sqrt{6}}{\sqrt{6+1}+\sqrt{6}}=2\times 6+1-2\sqrt{6}\sqrt{6+1}=13-2\sqrt{42}$.

24)

$$\frac{1}{\sqrt{a}+\sqrt{a+1}}\times\frac{\sqrt{a}-\sqrt{a+1}}{\sqrt{a}-\sqrt{a+1}}=\frac{\sqrt{a}-\sqrt{a+1}}{a-(a+1)}=\sqrt{a+1}-\sqrt{a}$$

$$\frac{1}{\sqrt{1}+\sqrt{2}}=\sqrt{2}-\sqrt{1},\quad \frac{1}{\sqrt{2}+\sqrt{3}}=\sqrt{3}-\sqrt{2},\ \ldots,\quad \frac{1}{\sqrt{99}+\sqrt{100}}=\sqrt{100}-\sqrt{99}$$

$$\begin{aligned}\frac{1}{\sqrt{1}+\sqrt{2}}+\frac{1}{\sqrt{2}+\sqrt{3}}+\cdots+\frac{1}{\sqrt{99}+\sqrt{100}} &= \sqrt{2}-\sqrt{1}+\sqrt{3}-\sqrt{2}+\sqrt{4}-\sqrt{3}\\ &\quad+\sqrt{5}-\sqrt{4}+\cdots+\sqrt{99}-\sqrt{98}+\sqrt{100}-\sqrt{99}\\ &= -\sqrt{1}+\sqrt{100}=-1+\sqrt{100}=9\end{aligned}$$

25)

If $a>b>0$ then $a^2>b^2$.

$$(\sqrt{2}+\sqrt{3})^2=2+2\sqrt{2}\sqrt{3}+3=5+2\sqrt{6}$$

$$(\sqrt{5})^2=5$$

So $5+2\sqrt{6}>5\Rightarrow(\sqrt{2}+\sqrt{3})^2>(\sqrt{5})^2\Rightarrow\sqrt{2}+\sqrt{3}>\sqrt{5}$ as req.

26)

$$\begin{aligned}
\frac{2}{\sqrt{5}+(\sqrt{2}+1)} \times \frac{\sqrt{5}-(\sqrt{2}+1)}{\sqrt{5}-(\sqrt{2}+1)} &= \frac{2\sqrt{5}-2\sqrt{2}-2}{5-(2+2\sqrt{2}+1)} \\
&= \frac{2\sqrt{5}-2\sqrt{2}-2}{2-2\sqrt{2}} \times \frac{2+2\sqrt{2}}{2+2\sqrt{2}} \\
&= \frac{4\sqrt{5}-4\sqrt{2}-4+4\sqrt{10}-8-4\sqrt{2}}{4-8} \\
&= \frac{-12-8\sqrt{2}+4\sqrt{5}+4\sqrt{10}}{-4} \\
&= 3+2\sqrt{2}-\sqrt{5}-\sqrt{10}
\end{aligned}$$